Band-1 Computational Experiment: De Bruijn–Newman constant Λ — literature ledger + Xi/Z and toy strip methods validation

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note does **not** claim the Riemann hypothesis (RH). It does **not** claim a new world-record bound on the de Bruijn–Newman constant Λ. It records the **peer-reviewed standing enclosure** $0 \le \Lambda \le 0.2$, reproduces one published **arithmetic gate** in the Platt–Trudgian / Polymath15 corollary chain, and runs **independent methods-validation** certificates (Hardy Z sign-change isolation; clearly labeled TOY strip zero checks). A non-peer-reviewed candidate $\Lambda \le 0.1787854$ is noted separately and **not** adopted as standing.

Abstract

For each real t, the de Bruijn–Newman functions H_t are heat-flow evolutions of the Riemann ξ-function. There exists a finite constant Λ such that all zeros of H_t are real precisely when $t \ge \Lambda$. Classically, RH $\Leftrightarrow \Lambda \le 0$. Rodgers–Tao proved $\Lambda \ge 0$. Polymath15 proved $\Lambda \le 0.22$; Platt–Trudgian, verifying RH to height $3 \cdot 10^{12}$, unlocked Polymath15's table row giving $\Lambda \le 0.2$.

Tonight's Band-1 probe:

1. locks the literature ledger $0 \le \Lambda \le 0.2$;
2. checks the numeric gate $3 \cdot 10^{12} > 2.51 \cdot 10^{12}$ (citation-faithful; **not** a re-verification of RH to that height);
3. isolates the first 30 critical-line zeros of ζ via Hardy Z sign changes (max $|t - t_{\text{Odlyzko}}| \approx 2.96 \cdot 10^{-7}$);
4. certifies, as **TOY / methods validation**, that Hermite He_{24} has 24 real zeros in a strip, and that a mean-zero Gaussian cosine sum has winding number matching its real-axis zero count with no off-axis near-zeros in a tested rectangle.

Problem statement

De Bruijn–Newman constant. Define

$$H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du,$$

with Φ the usual super-exponentially decaying even kernel built from the Jacobi theta / Riemann ξ data. Then $\Lambda \in \mathbb{R}$ is characterized by: all zeros of H_t are real iff $t \geq \Lambda$.

- **Open target (Band 1):** shrink the enclosure of Λ (equivalently, probe how "barely true" RH can be if $\Lambda = 0$).
- **Finite checks (this note):** literature ledger + small reproducible lemmas / toys.

Literature ledger (peer-reviewed standing bounds)

Bound	Value	Source
Lower	$\Lambda \geq 0$	Rodgers–Tao, *Forum Math. Pi* **8** (2020), e6; arXiv:1801.05914
Upper	$\Lambda \leq 0.2$	Platt–Trudgian, *Bull. London Math. Soc.* **53** (2021) 792–797 (arXiv:2004.09765), using Polymath15 Table 1
Prior upper	$\Lambda \leq 0.22$	Polymath15, *Res. Math. Sci.* **6** (2019), Paper 31; arXiv:1904.12438
Prior upper	$\Lambda < 1/2$	Ki–Kim–Lee (2008)
Classical	RH $\Leftrightarrow \Lambda \leq 0$	de Bruijn–Newman

Standing enclosure adopted here (2026-09):

$$\boxed{0 \leq \Lambda \leq 0.2}.$$

Candidate (not adopted)

Gomila (2025–2026 computer-assisted candidate audit) proposes $\Lambda \leq 893927/5000000 = 0.1787854$. The author's repository states the result is **not peer-reviewed**. This lab **does not** treat 0.1787854 as the standing upper bound.

Historical lower-bound path (context)

Year	Lower bound on Λ
1987	-50

Year	Lower bound on Λ
1990	-5
1991	-0.0991
2011	$-1.1 \cdot 10^{-11}$ (Saouter–Gourdon–Demichel)
2018/2020	0 (Rodgers–Tao)

Method

A. Citation-faithful arithmetic gate (Platt–Trudgian Corollary 2)

Polymath15 Table 1 implies: if RH is verified to height $H > 2.51 \cdot 10^{12}$, then $\Lambda \leq 0.2$. Platt–Trudgian prove RH up to $H = 3 \cdot 10^{12}$. We recompute only the comparison

$$3 \cdot 10^{12} \; > \; 2.51 \cdot 10^{12},$$

which holds. **We do not re-verify RH to height $3 \cdot 10^{12}$.**

B. Hardy Z sign-change isolation (Xi-style infrastructure)

The Hardy function $Z(t) = e^{i\theta(t)}\zeta(1/2 + it)$ is real for real t; its real zeros are the critical-line zeros of ζ. We:

1. evaluate Z via `scipy.special.zeta` and Riemann–Siegel θ from $\log\Gamma$;
2. run a coarse grid on $(0, 50]$ (step 0.05) and count sign changes;
3. around each of the first 30 Odlyzko reference zeros, locate a sign-changing bracket and bisect.

C. TOY strip certificates (clearly labeled)

TOY-1 (Hermite). Probabilists' Hermite polynomials He_n are heat-/Fourier-related. Classically all zeros are real. We re-verify He_{24} via companion-matrix roots + strip near-zero scan.

TOY-2 (cosine sum). $f(z) = \sum_{k=1}^{N} e^{-\alpha k^2} \cos(kz)$ (mean-zero Gaussian weights). Real-axis sign changes + boundary winding + off-axis near-zero scan on a rectangle.

These toys validate strip-search tooling **analogous in spirit** to H_t zero location. They **do not** bound Λ.

Method limits

- No full Polymath15 Fourier / barrier computation.
- No re-proof of Rodgers–Tao.
- No claim that numerical Z-isolation proves RH.
- TOY certificates are finite numerical checks with stated tolerances, not formal interval-arithmetic proofs of Λ-bounds.
- SciPy complex ζ is used at moderate height ($t \leq 102$); not a substitute for rigorous ball arithmetic at 10^{12}.

Experiment design

Parameter	Choice
Probe	De Bruijn–Newman ledger + methods validation
Slug	`de_bruijn_newman`
Host	Linux box, Python 3.13 + NumPy/SciPy
Success criterion	Ledger locked; gate PASS; 30/30 Z-isolations; both toys PASS; no overclaim
Wall budget	$\ll 1$ minute

Code: `/workspace/experiments/band1/de_bruijn_newman/verify_dbn.py`
Lab mirror: `/workspace/mission-control/labs/de_bruijn_newman/`
Raw JSON: `results.json`

Results (numbers!)

A. Arithmetic gate

- $H_{\text{Platt–Trudgian}} = 3 \cdot 10^{12}$
- $H_{\text{required (Polymath15 row for 0.2)}} = 2.51 \cdot 10^{12}$
- Comparison holds: **PASS**
- Conclusion under published theorems: $\Lambda \leq 0.2$ (not recomputed from scratch)

B. Hardy Z / coarse grid

Quantity	Value
Coarse grid T	50
Step	0.05
Samples	1000
Sign changes on $(0, 50]$	10 (matches 10 Odlyzko zeros in range)
Wall	≈ 0.006–0.009 s

C. Isolation of first 30 Odlyzko zeros

Quantity	Value
Targets	30
Isolated	30
Failures	0
$\max \lVert t - t_{\text{Odlyzko}} \rVert$	$2.962 \cdot 10^{-7}$
$\max \lVert Z(t_{\text{isolated}}) \rVert$	$1.307 \cdot 10^{-10}$

Quantity	Value
Wall	≈ 0.012 s
Pass	**PASS**

Sample (first five):

n	Odlyzko t	Isolated t	$\|Z\|$
1	14.1347251417	14.1347251417	$\sim 10^{-11}$
2	21.0220396388	21.0220396388	$\sim 10^{-11}$
3	25.0108575801	25.0108575801	$\sim 10^{-11}$
4	30.4248761259	30.4248761259	$\sim 10^{-11}$
5	32.9350615877	32.9350615877	$\sim 10^{-10}$

D. TOY-1 — Hermite He_{24}

Quantity	Value
Degree n	24
Companion roots	24
max $\|\mathrm{Im}(\mathrm{root})\|$	0
Real-axis sign changes on $[-X, X]$	24
Off-axis near-zeros in strip	0
Pass	**PASS**
Wall	≈ 0.014 s

E. TOY-2 — mean-zero Gaussian cosine sum

Quantity	Value
α	0.05
N	30
Rectangle	$\|Re\| \leq 40$, $\|Im\| \leq 0.5$
Real-axis zeros isolated	26
Boundary winding	≈ 26.000
Off-axis near-zeros	0
Pass	**PASS**
Wall	≈ 0.058 s

Overall

overall_pass = true (ledger + gate + Z-isolation + both toys).

Explicitly not claimed / not done

- Riemann hypothesis
- Any new upper bound on Λ tighter than 0.2
- Adoption of the non-peer-reviewed 0.1787854 candidate as standing
- Re-verification of RH to height $3 \cdot 10^{12}$
- Full Polymath15 barrier / mollifier pipeline
- Clay, ToE, or prize stamps

Reproducibility

```
python3 /workspace/experiments/band1/de_bruijn_newman/verify_dbn.py
```

Log: `/workspace/experiments/logs/2026-09-11_de_bruijn_newman.log`

Key numbers (return block)

- Literature: $0 \leq \Lambda \leq 0.2$
- Gate: $3 \cdot 10^{12} > 2.51 \cdot 10^{12}$ → PASS
- Z-zeros isolated: $30/30$, max abs err vs Odlyzko $2.962 \cdot 10^{-7}$
- TOY Hermite He_{24}: max $\|\mathrm{Im}\| = 0$, 24 sign changes, PASS
- TOY cosine: 26 real zeros, winding 26, off-axis 0, PASS
- PDF: `/workspace/mission-control/papers/2026-09-11_Band1_Computational_Experiment_de`

References

1. B. Rodgers, T. Tao, The de Bruijn–Newman constant is non-negative, *Forum Math. Pi* **8** (2020), e6.
2. D. H. J. Polymath, Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant, *Res. Math. Sci.* **6** (2019), Paper 31; arXiv:1904.12438.
3. D. Platt, T. Trudgian, The Riemann hypothesis is true up to $3 \cdot 10^{12}$, *Bull. London Math. Soc.* **53** (2021) 792–797; arXiv:2004.09765.
4. H. Ki, Y. Kim, J. Lee, On the de Bruijn–Newman constant, *Adv. Math.* (2008) line: $\Lambda < 1/2$.
5. A. M. Odlyzko, tables of zeros of the Riemann zeta function (reference targets for Z-isolation).
6. J. Gomila, candidate $\Lambda \leq 0.1787854$ audit (not peer-reviewed; not adopted here).